

SHI JIN (Chris)

sjin11@stevens.edu | +1 (650) 639-6585 | United States
github.com/Chris-Jin02 | linkedin.com/in/sjin0907

RESEARCH PROFILE

M.S. Computer Science student (GPA: 4.00/4.00) developing reproducible, data-intensive machine-learning systems across multimodal sensing, medical imaging, information retrieval, and server diagnostics. Strengths include model optimization through controlled preprocessing, parameter validation, baseline iteration, error analysis, and repeatable evaluation. Interested in data-to-decision research that integrates Earth-system analytics, interpretable AI, and optimization to improve the resilience of agricultural, forest, and critical infrastructure systems under environmental uncertainty.

EDUCATION

Stevens Institute of Technology | M.S. in Computer Science | GPA: 4.00/4.00

Expected May 2027

Stevens Master's Scholarship Hoboken, NJ, USA

- Selected coursework (all A): Mathematical Foundations of Machine Learning; Data Structures and Algorithms; Text Mining and Information Retrieval; Academic Communication.

Ulster University | B.Eng. in Engineering (Joint Program) Sep 2020 - Jun 2024

Belfast, United Kingdom

Shaanxi University of Science and Technology | B.Eng. in Computer Science and Technology

Sep 2020 - Jun 2024

Xi'an, China

- Highlighted coursework: Advanced Mathematics (96), Probability Theory (90), System Security (96), Databases (87).

RESEARCH INTERESTS

- Earth-human systems modeling, remote sensing, hydroclimatic extremes, and resilience of agriculture, forests, and critical infrastructure.
- Hybrid process-ML modeling, observation-constrained residual learning, model calibration, bias correction, uncertainty quantification, and interpretable AI.
- Stochastic and robust optimization, resource allocation, and decision-support systems under environmental uncertainty.

TECHNICAL SKILLS

Modeling & optimization: PyTorch, YOLO, EfficientNet, LSTM, diffusion-model workflows; baseline selection, controlled preprocessing, parameter validation, augmentation, dimensionality reduction, error analysis, and reproducible evaluation.

Data & computing: Python, Java, C; OpenCV, PyDICOM; web crawling, regex/ETL, quality control, MySQL, MongoDB; Matplotlib, ECharts.

Systems engineering: Django, Flask, REST-style APIs, AJAX; multimodal alignment, data/API contracts, debugging, root-cause analysis, privacy-aware data handling.

RESEARCH EXPERIENCE

Research Developer | Standardized Modeling of the Sacrum Using Diffusion Models

Jul 2023 - Jun 2024

Supervisor: Prof. Yong Qi

Xi'an, China

- Built a modular DICOM-to-CT preprocessing and quality-control pipeline with missing-slice checks, noise handling, and pixel-anomaly detection, producing consistent inputs for diffusion-model experiments.
- Optimized experimental stability through parameter validation and controlled comparisons of preprocessing, augmentation, and dimensionality-reduction configurations.
- Collaborated with Xi'an Red Cross Hospital under privacy constraints, applying anonymization and controlled data handling throughout the workflow.

Research Assistant | Digital Twin and Smart Substation (Computer Vision / Robotics)

Nov 2021 - Feb 2022

Supervisor: Prof. Yong Qi

Xi'an, China

- Integrated infrared thermography and LiDAR data for substation inspection; performed SLAM-based registration to improve map, thermal, and point-cloud consistency for downstream modeling.
- Created training-ready labels with standardized device, bounding-box, and temperature metadata, then iterated YOLO and EfficientNet baselines using stable preprocessing and repeatable training runs.

Software Engineering Intern (Project-based) | Xi'an iSoftStone Network Technology Co., Ltd.

May 2023 - Jun 2023

Xi'an, China

- Built an end-to-end scholarly-data system spanning ingestion, cleaning, storage, retrieval, and visualization; collected and normalized 350K arXiv papers with Python and regex-based ETL.
- Developed Django retrieval APIs and AJAX/ECharts dashboards; introduced explicit schemas and serialization layers to eliminate cross-stage format mismatches.
- Implemented an LSTM baseline in PyTorch and standardized training/evaluation to support controlled, repeatable model comparisons.

PROPOSED DOCTORAL RESEARCH

Observation-Constrained Diagnosis and Correction of Ecosystem Model Biases During Flash Droughts Proposed

- Diagnose phase-specific, structured residuals in gross primary productivity (GPP), actual evapotranspiration (AET), and water-use efficiency (WUE) by aligning flux-tower observations, land-model outputs, and meteorological/environmental drivers.
- Develop interpretable and temporal residual-learning models that correct GPP and AET while preserving the process-based model as the scientific backbone; optimize model configuration using event-aware validation, error decomposition, and reproducible ablation studies.
- Evaluate not only RMSE and bias, but also event-phase accuracy, carbon-water coupling, and post-drought recovery timing; quantify where corrections generalize across events, sites, and ecosystem types.
- Use soil moisture, vapor-pressure deficit, temperature, precipitation, vegetation state, and available plant-hydraulic indicators to identify when model errors are systematic and which environmental controls explain them.
- Extend corrected predictions and uncertainty estimates into data-to-decision frameworks for robust or stochastic irrigation, resource-allocation, and resilience planning under compound drought and flood risk.

INDUSTRY EXPERIENCE

Server Test Engineer Intern | Quanta Nashville Jun 2026 - Present

La Vergne, TN, USA

- Execute hardware/software diagnostics, isolate component failures, complete repair verification and retest, and validate restored functionality against product specifications.
- Maintain traceable QMS test and defect records and communicate reproducible failure evidence to support root-cause analysis, recurring-pattern identification, and continuous improvement.
- Coordinate with Test Operations and production teams in a high-throughput environment while following process, labeling, documentation, and safety requirements.

SELECTED TECHNICAL PROJECTS

Scholarly Data Platform | ETL, Retrieval, and Interactive Analytics Python, Django, MongoDB

- Designed stage-wise data contracts and ingestion-time schema checks at 350K-record scale, reducing integration failures and enabling maintainable retrieval and visualization workflows.
- Built retrieval endpoints and AJAX/ECharts refresh flows that connected normalized data to interactive analytical views.

Medical Imaging Pipeline | Reproducible Diffusion-Model Workflow PyDICOM, OpenCV, PyTorch

- Modularized CT preprocessing and quality control and enforced parameter checks so diffusion-model iterations could be compared on consistent, privacy-aware inputs.
- Evaluated controlled preprocessing and feature-reduction alternatives as stability-focused model-optimization steps.

Multimodal Inspection Workflow | Alignment and Annotation Readiness Python, LabelMe

- Standardized cross-sensor registration, labeling conventions, and metadata structures to improve robustness across collection sessions and downstream vision models.
- Prepared aligned infrared and LiDAR observations for repeatable object-detection and classification experiments.

HONORS & AWARDS

- **Stevens Master's Scholarship.**
- 1st Prize, China National College Computer Competition (Northwest Regional), 2022.
- 3rd Prize, 15th China National Undergraduate Computer Design Contest, Jul-Aug 2022.
- Successful Participant, Mathematical Contest in Modeling (MCM), 2022.
- Excellence Award, China AI+ Innovation & Entrepreneurship Competition (Industrial Internet), Dec 2021.
- 1st Prize, Intelligent Robot Innovation Design, National Robot Championship, Oct 2021.